**Joel Alex John**
+91-8089232301 | Email | LinkedIn | GitHub | Medium

## Summary

A result-driven AI and ML Engineer with hands-on experience in Computer Vision, NLP and Agentic AI. Skilled in solving complex problems by providing innovative solutions and approaches. Continuously exploring and learning new techniques and technologies to be able to solve real-world challenges.

## Education

**Christ University** Bengaluru, IN
*Bachelor of Technology in Artificial Intelligence & Machine Learning; GPA: 9.3* *2022 – Present*

## Experience

**AI/ML Development Intern** December 2025 – January 2026
*Erytos Technologies Pvt. Ltd.*

- Built a multi-modal agent that analyses mmWave radar vitals using Random Forest and autonomously triggers speech interaction and emergency escalation via an LLM-orchestrated control loop.
- Integrated Vosk based STT/TTs for real time voice interaction and emergency contact actions without user prompts.

**Artificial Intelligence Intern** May 2024 – July 2024
*Intel Unnati*

- Developed and fine-tuned an ESPCN model for pixelation removal which achieved PSNR of 32 and SSIM of 81% on the Div2k dataset.
- Built a custom data pipeline processing 10K+ images, integrated artifact detection and removal models, and reduced training time by 77%.

**Artificial Intelligence Intern** April 2024 – May 2024
*TechnoHacks EduTech*

- Developed a CNN using the Mediapipe framework for emotion detection which achieved an accuracy of 81% on a set of over 2000 images.
- Generated over 500 anime style face images for dataset augmentation.

## Projects

**Reddit Persona Builder** | *Agentic AI, Python*

- Developed a tool in Python which scrapes up to 100 user posts and comments from Reddit's public JSON API, processes the text and puts together a profile for personality analysis.
- Integrated into Google Gemini 1.5 Flash which puts out structured JSON results with persona profiles, interests, tone and behavioral data.
- Implemented a solution for text parsing and API pagination which in turn gives us accurate data retrieval, response formatting and automated report generation in a human readable format.

**Stock Price Predictor using LSTM and Transformer** | *LSTM, FinBERT, NIG, GBM*

- Integrated into neural networks the NIG and GBM models which performed with an $R^2$ of up to 0.98.
- Combined with FinBERT using the Hugging Face API for precise predictions.

**Web Research Agent** | *Agentic AI, Python*

- Developed a Web Research Agent that uses SERP API search, BeautifulSoup scraping, and Gemini 1.5 Pro LLM for automated content analysis and synthesis.
- Implemented a smart query expansion, caching, rate limiting and fallback strategies which in turn improved relevance, reduced API costs and ensured high availability.
- Built a Gradio based front end which allows users to put in their research questions, get back AI generated summaries, and provide source links.

## Certifications

AWS Certified Cloud Practitioner – **AWS**
Data Science For Engineers – **NPTEL**
Google Cloud Big Data and Machine Learning Fundamentals – **GCP**
Introduction to AI and Machine Learning on Google Cloud – **GCP**
How Google Does Machine Learning – **GCP**
Advanced Computer Vision with TensorFlow – **Coursera**
Generative Deep Learning with TensorFlow – **Coursera**
Master Computer Vision™ OpenCV4 in Python with Deep Learning – **Udemy**
Face Recognition using Python Language – **Udemy**

## Achievements

Secured 2nd place in Academic Performance for the 2023-2024 academic year in the Artificial Intelligence Machine Learning program at Christ University.

## Technical Skills

**Languages**: Python, C/C++, R, SQL
**Frameworks & Libraries**: TensorFlow, Scikit-learn, Keras, OpenCV, Mediapipe, NLTK, Pandas, NumPy, Matplotlib, Seaborn, Hugging Face
**Developer Tools**: Git, Google Cloud Platform, Visual Studio Code, Google Colab